## 7. Polar coordinates

### What students need to learn:

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|---|---|
| Polar coordinates $(r, \theta)$, $r \geqslant 0$. | The sketching of curves such as $\theta = \alpha$, $r = p \sec(\alpha - \theta)$, $r = a$, $r = 2a \cos\theta$, $r = k\theta$, $r = a(1 \pm \cos\theta)$, $r = a(3 + 2\cos\theta)$, $r = a \cos 2\theta$ and $r^2 = a^2 \cos 2\theta$ may be set. |
| Use of the formula $\frac{1}{2}\int_{\alpha}^{\beta} r^2 \, \mathrm{d}\theta$ for area. | The ability to find tangents parallel to, or at right angles to, the initial line is expected. |